

Mathematics Stage 3 -- Year B -- Unit 23 Length & Time

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-GM-02	Selects and uses the appropriate unit and device to measure lengths and distances including perimeters.
MA3-NSM-02	Measures and compares duration, using 12- and 24-hour time.
MA3-2DS-02	Selects and uses the appropriate unit to calculate areas, including areas of rectangles.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- Applying	Green -- Extension
Measure length & perimeter (GM-A), decimal length (GM-B), 24-hour time (NSM-A) Use a variety of measuring devices to measure lengths and distances; use metres and kilometres (GM-A). Calculate perimeters of common 2D shapes, including squares, rectangles and triangles (GM-A). Recognise the equivalence of whole-number and decimal length; interpret and record lengths in decimal notation (GM-B). Compare 12- and 24-hour time and convert between them; read time using 24-hour language (NSM-A).	Convert units, perimeter & area, elapsed time & timetables (applying) Use place value to convert between metres and kilometres; convert to the same unit to compare lengths (GM-B). Use efficient strategies for perimeter of a large rectangle; determine which side lengths are needed (GM-A). Recognise same-perimeter rectangles can differ; compare perimeters of same-area rectangles; calculate area of rectangles (GM-A/B, 2DS-A). Calculate elapsed time using bridging; round to the nearest minute or hour (NSM-B). Read and use 12- and 24-hour timetables; solve duration problems (NSM-A/B).	Extended -- Stage 4 Area, Rates & Distance-Time Graphs Apply the formula for the area of a triangle ($A = \frac{1}{2}bh$); find areas of composite figures (MA4-ARE-C-01). Solve rate problems, including speed = distance / time; convert between units for rates (MA4-RAT-C-01). Interpret distance-time graphs and describe a journey (MA4-RAT-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students record lengths in decimal notation, read 24-hour time and timetables, and explain how they found a perimeter or duration.
Understanding and fluency	Students measure with metres and kilometres, calculate perimeters, convert units of length, and calculate elapsed time.
Reasoning	Students reason that same-perimeter rectangles can differ in area, choose which side lengths are needed, and check durations are reasonable.
Problem solving	Students solve perimeter, conversion, timetable and duration problems, and (Stage 4) triangle/composite area, rate and distance-time-graph problems.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Studen ts who need further consol idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 23 | NESA Mathematics K-10 Syllabus (2022)

Unit 23 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Measure length with metres and kilometres.	GM-A: use a variety of measuring devices; use metres and kilometres for length and distances MA3-GM-02 Year A	Mini Lesson 1		
R2	Calculate perimeter of squares, rectangles and triangles.	GM-A: calculate perimeters of common 2D shapes, including squares, rectangles and triangles MA3-GM-02 Year A	Mini Lesson 2		
R3	Record and interpret lengths in decimal notation.	GM-B: equivalence of whole-number and decimal length; interpret and record lengths in decimal notation MA3-GM-02 Year B	Mini Lesson 3		
R4	Read and convert 12- and 24-hour time.	NSM-A: compare 12- and 24-hour time and convert; read time using 24-hour language MA3-NSM-02 Year A	Mini Lesson 4		
Y1	Convert between metric units of length.	GM-B: convert between metres and kilometres using place value; convert to the same unit to compare MA3-GM-02 Year B	Mini Lesson 5		
Y2	Find side lengths needed and calculate perimeter.	GM-A: efficient perimeter of a large rectangle; determine which side lengths are needed MA3-GM-02 Year A	Mini Lesson 6		
Y3	Compare perimeter and area of rectangles.	GM-A/B & 2DS-A: same-perimeter rectangles differ; compare same-area perimeters; area of rectangles MA3-GM-02, MA3-2DS-02 Year A/B	Mini Lesson 7		
Y4	Calculate elapsed time.	NSM-B: elapsed time from start/finish using bridging; round to the nearest minute or hour MA3-NSM-02 Year B	Mini Lesson 8		
Y5	Read timetables and solve duration problems.	NSM-A/B: read and use 12-/24-hour timetables; solve duration problems MA3-NSM-02 Year A/B	Mini Lesson 9		
G1	Area and composite area.	Stage 4 ARE: area of a triangle ($A = 1/2 b h$); area of composite figures MA4-ARE-C-01 Stage 4 (Core)	Mini Lesson 10		
G2	Distance, speed and time (rates).	Stage 4 RAT: solve rate problems incl. speed = distance / time; convert units for rates MA4-RAT-C-01 Stage 4 (Core)	Mini Lesson 11		
G3	Interpret distance-time graphs.	Stage 4 RAT: interpret distance-time graphs; describe a journey MA4-RAT-C-01 Stage 4 (Core)	Mini Lesson 12		

Unit 23 -- Mini Lesson Sequence

Length & Time | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-4

Yellow band -- On-level -- Mini Lessons 5-9

Green band -- Extension -- Mini Lessons 10-12

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 23. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Measuring length with metres and kilometres [HANDS-ON]

R1 | Red band | DoE Lesson 8 (adapted)

Duration: 25-30 min | **Outcomes:** R1 | Syllabus: MA3-GM-02, MAO-WM-01 | GM-A: measuring devices; metres and kilometres

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Measuring distance and solving duration problems | Key resources: Rulers, tape measures, trundle wheel

Physical materials needed: rulers, tape measures, trundle wheel

Learning intention

Students are learning to:
choose a suitable device and unit to measure lengths and distances

Success criteria

Students can:

- * I can choose the right measuring device for a length or distance.
- * I can use metres for lengths and kilometres for longer distances.
- * I know 1 kilometre is 1000 metres.

Mini masterclass structure (25-30 min)

Activate (4 min): Would you measure the classroom in metres or kilometres? What about the trip to the next town?

Model (10 min): Match the device to the job: ruler/tape for metres, trundle wheel for the playground, kilometres for road distances. Show 1 km = 1000 m.

Guided practice (10 min): Students measure several lengths with suitable devices and record them in metres or kilometres.

Check (4 min): Exit question: which unit (m or km) would you use for the length of a pencil, a pool and a highway?

Key vocabulary

length, distance, metre, kilometre, measuring device, ruler, tape measure, trundle wheel

Differentiation

Support: Provide a chart linking each device to the kind of length it measures.
Extension: Estimate a distance first, then measure, and compare estimate to result.

Assessment

Students choose units and devices to measure length — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Calculating perimeter

R2 | Red band | DoE Lesson 3 (adapted)

Duration: 25 min **Outcomes:** R2 Syllabus: MA3-GM-02, MAO-WM-01 | GM-A: perimeter of squares, rectangles, triangles

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Calculating perimeter | Key resources: Grid paper, 2D shape cards

Learning intention		Success criteria	
Students are learning to: calculate the perimeter of squares, rectangles and triangles		Students can: * I know perimeter is the total distance around the outside of a shape. * I can add the side lengths to find the perimeter. * I can use the shape's properties (e.g. a square has 4 equal sides).	
Mini masterclass structure (25 min)			
Activate (3 min): How far is it around the edge of this rectangle? What do we call that distance? Model (10 min): Perimeter = add all sides. Rectangle 5 m by 3 m: 5 + 3 + 5 + 3 = 16 m. Square side 4 cm: 4 × 4 = 16 cm. Triangle: add the 3 sides. Guided practice (8 min): Students calculate perimeters of squares, rectangles and triangles. Check (4 min): Exit question: a rectangle is 6 cm by 2 cm. What is its perimeter?			
Key vocabulary		Differentiation	
perimeter, side, length, square, rectangle, triangle, total, distance around		Support: Let students trace the outline and count grid squares. Extension: Find a missing side length given the perimeter and the other sides.	
Assessment		Teaching notes (annotate here)	
Students calculate perimeter — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 3 Lengths in decimal notation

R3 | Red band | DoE Lesson 1 & 2 (adapted)

Duration: 25-30 min | **Outcomes: R3** | Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: decimal notation for length

Linked to: DoE Lesson 1 & 2 (adapted) -- Lesson 1 -- Converting between metres and kilometres | Key resources: Metre tapes, place value charts

Learning intention		Success criteria
Students are learning to: record and interpret lengths in decimal notation		Students can: * I can write a length in decimal notation (e.g. 1500 m = 1.5 km). * I know the equivalence of whole-number and decimal length. * I can interpret a decimal length and say what it means.
Mini masterclass structure (25-30 min)		
Activate (4 min): A sign says 1.5 km. How many metres is that? Model (10 min): 1000 m = 1 km, so 1500 m = 1.5 km. 250 cm = 2.5 m. The decimal shows part of the larger unit. Read 2.75 km as 2 km and 750 m. Guided practice (10 min): Students write and interpret lengths in decimal notation. Check (4 min): Exit question: write 1750 m in kilometres using a decimal.		
Key vocabulary		Differentiation
length, decimal notation, metre, kilometre, equivalent, interpret, record		Support: Use a place value chart aligning metres with the decimal places of kilometres. Extension: Order a set of lengths given in mixed units (m and km) by converting first.
Assessment		Teaching notes (annotate here)
Students record and interpret decimal lengths — links to R3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Reading and converting 12- and 24-hour time

R4 | Red band | DoE Lesson 7 (adapted)

Duration: 25-30 min

Outcomes: R4

Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-A: 24-hour time; convert 12<->24

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Interpreting and using timetables | Key resources: Analog/digital clocks, 24-hour time cards

Learning intention

Students are learning to:
read 24-hour time and convert between 12- and 24-hour time

Success criteria

Students can:

- * I can read time written in 24-hour notation (e.g. 14:30).
- * I can convert 12-hour time to 24-hour time and back.
- * I know that after 12 noon you add 12 to the hour for 24-hour time.

Mini masterclass structure (25-30 min)

Activate (3 min): A bus leaves at 15:00. What time is that on a 12-hour clock?

Model (10 min): Up to noon, the hour is the same (9:00 am = 09:00). After noon, add 12: 3:00 pm = 15:00. Midnight = 00:00.

Guided practice (10 min): Students convert times both ways between 12- and 24-hour notation.

Check (4 min): Exit question: write 7:45 pm in 24-hour time, and 06:20 in 12-hour time.

Key vocabulary

12-hour time, 24-hour time, am, pm, noon, midnight, convert

Differentiation

Support: Provide a 12-to-24-hour conversion strip for the afternoon hours.
Extension: Convert times that cross midnight (e.g. 11:50 pm) and explain.

Assessment

Students read and convert time — links to R4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Converting between metric units of length

Y1 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: convert m $\leftrightarrow$ km; compare lengths

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Converting and comparing units of length | Key resources: Place value charts, conversion cards

Learning intention		Success criteria
Students are learning to: convert between metres and kilometres and compare lengths in the same unit		Students can: * I can convert metres to kilometres and back using place value ($\times$ or $\div$ 1000). * I can convert lengths to the same unit before comparing them. * I know a larger unit needs fewer of them to measure the same length.
Mini masterclass structure (30 min)		
Activate (4 min): Which is longer, 1300 m or 1.2 km? Model (12 min): 1000 m = 1 km, so divide by 1000 to go m $\rightarrow$ km (1300 m = 1.3 km) and multiply by 1000 to go km $\rightarrow$ m. Convert to the same unit to compare: 1300 m vs 1200 m $\rightarrow$ 1300 m is longer. Guided practice (10 min): Students convert and compare lengths given in mixed units. Check (4 min): Exit question: order 1.4 km, 1350 m and 1.25 km from shortest to longest.		
Key vocabulary		Differentiation
convert, metre, kilometre, place value, compare, unit, equivalent		Support: Provide a m $\leftrightarrow$ km conversion chart in multiples of 100 m. Extension: Convert and add lengths in mixed units, giving the total in kilometres.
Assessment		Teaching notes (annotate here)
Students convert and compare lengths — links to Y1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Side lengths needed for perimeter

Y2 | Yellow band | DoE Lesson 4 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-GM-02, MAO-WM-01 | GM-A: efficient perimeter; side lengths needed

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Determining the side lengths needed to calculate perimeter | Key resources: Shape cards with some sides labelled

Learning intention		Success criteria
Students are learning to: work out which side lengths are needed and find the perimeter efficiently		Students can: * I can use a shape's properties to find unlabelled side lengths. * I can find the perimeter of a large rectangle efficiently (e.g. $2 \times (\text{length} + \text{width})$). * I can decide which measurements I actually need.
Mini masterclass structure (30 min)		
Activate (4 min): A rectangle shows only its length and width. Do you need all four sides labelled? Model (12 min): For a rectangle you only need length and width: perimeter = $2 \times (l + w)$. A 25 m by 10 m field: $2 \times (25 + 10) = 70$ m. Use opposite sides being equal to fill gaps. Guided practice (10 min): Students find missing side lengths and calculate perimeters efficiently. Check (4 min): Exit question: a rectangle is 12 m by 8 m. Use $2 \times (l + w)$ to find the perimeter.		
Key vocabulary		Differentiation
perimeter, side length, rectangle, opposite sides, efficient, formula, property		Support: Label opposite sides in the same colour to show they are equal. Extension: Find the perimeter of an L-shape by working out the missing sides.
Assessment		Teaching notes (annotate here)
Students find side lengths and perimeter — links to Y2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Comparing perimeter and area of rectangles [HANDS-ON]

Y3 | Yellow band | DoE Lesson 5 (adapted)

Duration: 30 min | **Outcomes:** Y3 | Syllabus: MA3-GM-02, MA3-2DS-02, MAO-WM-01 | GM-A/B & 2DS-A: perimeter vs area

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Comparing perimeter and area of shapes | Key resources: Grid paper, square tiles

Physical materials needed: grid paper, square tiles

Learning intention		Success criteria	
Students are learning to: compare the perimeter and area of rectangles		Students can: * I can calculate the area of a rectangle (length × width). * I know rectangles with the same perimeter can have different areas. * I can compare the perimeters of rectangles that have the same area.	
Mini masterclass structure (30 min)			
Activate (4 min): Two rectangles both have a perimeter of 12 cm. Must they have the same area? Model (12 min): Perimeter 12 cm: a 5 × 1 rectangle has area 5 cm ² ; a 3 × 3 has area 9 cm ² . Same perimeter, different area. Then compare two same-area rectangles' perimeters. Guided practice (10 min): Students build rectangles on grid paper and compare perimeter and area. Check (4 min): Exit question: a rectangle is 6 cm by 2 cm. Find its area and its perimeter.			
Key vocabulary		Differentiation	
perimeter, area, rectangle, length, width, square centimetre, dimensions		Support: Use square tiles so area (squares inside) and perimeter (edge) are concrete. Extension: Find all rectangles with whole-number sides and area 24 cm ² ; compare their perimeters.	
Assessment		Teaching notes (annotate here)	
Students compare perimeter and area — links to Y3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 8 Calculating elapsed time

Y4 | Yellow band | DoE Lesson 6 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-B: elapsed time; bridging; rounding

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Calculating elapsed time | Key resources: Timelines, clock faces

Learning intention		Success criteria
Students are learning to: calculate elapsed time using bridging and round to the nearest minute or hour		Students can: * I can find the elapsed time between a start and finish time. * I can use bridging (jump to the next hour) to add or subtract time. * I can round a time answer to the nearest minute or hour.
Mini masterclass structure (30 min)		
Activate (4 min): A film starts at 2:40 pm and ends at 4:15 pm. How long is it? Model (12 min): Bridge to the next hour: 2:40 → 3:00 is 20 min, 3:00 → 4:15 is 1 h 15 min, total 1 h 35 min. Show the same on a timeline. Guided practice (10 min): Students calculate elapsed times using bridging and a timeline. Check (4 min): Exit question: from 9:50 am to 11:20 am, how much time has passed?		
Key vocabulary		Differentiation
elapsed time, duration, start, finish, bridging, timeline, round, minute, hour		Support: Provide a blank timeline to mark the hour jumps. Extension: Find a finish time given a start time and a duration that crosses the hour.
Assessment		Teaching notes (annotate here)
Students calculate elapsed time — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Reading timetables and solving duration problems

Y5 | Yellow band | DoE Lesson 7 (adapted)

Duration: 30 min | **Outcomes:** Y5 | Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-A/B: timetables; duration

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Interpreting and using timetables | Key resources: Bus/train timetables (12- and 24-hour)

Learning intention		Success criteria	
Students are learning to: read 12- and 24-hour timetables and solve duration problems		Students can: * I can read a timetable and find the time of a given service. * I can work out how long a journey takes from a timetable. * I can solve duration problems using 12- and 24-hour time.	
Mini masterclass structure (30 min)			
Activate (4 min): A bus leaves at 09:15 and arrives at 09:50. How long is the trip? Model (12 min): Read down a column for one service; across a row for one stop. Find departure and arrival, then calculate the duration (09:15 to 09:50 = 35 min). Guided practice (10 min): Students answer questions from a real timetable, including journey durations. Check (4 min): Exit question: using the timetable, how long is the 14:10 service to its last stop at 14:55?			
Key vocabulary		Differentiation	
timetable, service, departure, arrival, duration, 24-hour time, journey		Support: Highlight the row and column being read. Extension: Plan a two-leg journey with a connection and find the total time.	
Assessment		Teaching notes (annotate here)	
Students read timetables and find durations — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 10 Area of triangles and composite shapes

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes: G1** | Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4: area of a triangle; composite area

Learning intention		Success criteria	
Students are learning to: find the area of a triangle and of composite shapes		Students can: * I can use $A = \frac{1}{2} \times \text{base} \times \text{height}$ to find the area of a triangle. * I can split a composite shape into rectangles and triangles. * I can add the parts to find the total area.	
Mini masterclass structure (30 min)			
Activate (4 min): A triangle has base 6 cm and height 4 cm. A rectangle with those sides is 24 cm ² . How does the triangle compare?			
Model (12 min): Triangle = half a rectangle: $A = \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$. For an L-shape, split into two rectangles, find each area, then add.			
Guided practice (10 min): Students find areas of triangles and composite shapes.			
Check (4 min): Exit question: find the area of a triangle with base 8 cm and height 5 cm.			
Key vocabulary		Differentiation	
area, triangle, base, height, composite, rectangle, formula, square centimetre		Support: Show the triangle as half of a drawn rectangle. Extension: Find the area of a composite shape made of a rectangle and a triangle.	
Assessment		Teaching notes (annotate here)	
Students find triangle and composite area — extends R2, Y2 and Y3.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 11 Distance, speed and time

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min **Outcomes:** G2 Syllabus: MA4-RAT-C-01, MAO-WM-01 | Stage 4: rates; speed = distance / time

Learning intention		Success criteria
Students are learning to: use the relationship between distance, speed and time to solve problems		Students can: * I know speed = distance / time. * I can find distance, speed or time when I know the other two. * I can use sensible units (km/h, m/s).
Mini masterclass structure (30 min)		
Activate (4 min): A car travels 120 km in 2 hours. How fast is it going? Model (12 min): speed = distance / time = $120 / 2 = 60$ km/h. distance = speed x time; time = distance / speed. Keep units matched. Guided practice (10 min): Students solve speed, distance and time problems. Check (4 min): Exit question: a runner covers 400 m in 80 s. What is their speed in m/s?		
Key vocabulary		Differentiation
speed, distance, time, rate, km/h, m/s, average speed		Support: Provide a distance-speed-time triangle to rearrange the formula. Extension: Convert a speed from km/h to m/s (or compare two speeds in different units).
Assessment		Teaching notes (annotate here)
Students solve speed-distance-time problems — extends R1, Y1, Y4 and Y5.		Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Interpreting distance-time graphs G3 Green band Original mini lesson (Stage 4 extension)		
Duration: 30 min	Outcomes: G3	Syllabus: MA4-RAT-C-01, MAO-WM-01 Stage 4: distance-time graphs
Learning intention		Success criteria
Students are learning to: read a distance-time graph and describe the journey		Students can: * I can read distance and time from a distance-time graph. * I know a steeper line means a faster speed, and a flat line means stopped. * I can describe a journey shown on a distance-time graph.
Mini masterclass structure (30 min)		
Activate (4 min): On a distance-time graph, what does a flat (horizontal) line mean? Model (12 min): Time is on the bottom, distance up the side. A rising line = moving away; steeper = faster. A flat line = stopped. A line back to zero = returning. Describe each part of a sample journey. Guided practice (10 min): Students read values and describe journeys from distance-time graphs. Check (4 min): Exit question: on a graph, a line is flat for 10 minutes. What is happening?		
Key vocabulary		Differentiation
distance-time graph, axis, gradient, steeper, flat, stopped, journey, speed		Support: Annotate a sample graph segment by segment. Extension: Sketch a distance-time graph for a described journey (out, stop, return).
Assessment		Teaching notes (annotate here)
Students interpret distance-time graphs — extends Y4 and Y5.		Taught as described / adapted / replaced -- note here:

Unit 23 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Measure length with metres and kilometres.		
Video	Metric Units of Length Convert mm, cm, m and km Math with Mr. J	https://www.youtube.com/watch?v=kOJFSH_Bn9U
Video	Common Metric Units of Length — kilometres, metres, centimetres and millimetres Math Mammoth	https://www.youtube.com/watch?v=CcrzyOqFmTU
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
R2 -- Calculate perimeter of squares, rectangles and triangles.		
Video	How to Find the Perimeter of a Rectangle Math with Mr. J	https://www.youtube.com/watch?v=teMDmMxy550
Video	How to Find Perimeter Math with Mr. J	https://www.youtube.com/watch?v=TSLc5p2VuYM
Worksheet (print)	Corbettmaths — Perimeter (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf
R3 -- Record and interpret lengths in decimal notation.		
Video	Meters and Centimeters — Converting m to cm and cm to m Math with Mr. J	https://www.youtube.com/watch?v=gALZRL4v2NM
Video	Metric Units of Length Convert mm, cm, m and km Math with Mr. J	https://www.youtube.com/watch?v=kOJFSH_Bn9U
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
R4 -- Read and convert 12- and 24-hour time.		
Video	The 24-Hour Clock Mathematics Grade 5 Periwinkle	https://www.youtube.com/watch?v=Lj4YxVZWLVU
Video	Understanding the 24-Hour Clock Smartli	https://www.youtube.com/watch?v=dQ5vIHA4L5s
Worksheet (print)	Corbettmaths Primary — Time (PDF)	https://corbettmathsprimary.com/wp-content/uploads/2018/07/time-pdf.pdf
Y1 -- Convert between metric units of length.		
Video	Metric Units of Length Convert mm, cm, m and km Math with Mr. J	https://www.youtube.com/watch?v=kOJFSH_Bn9U
Video	Meters and Centimeters — Converting m to cm and cm to m Math with Mr. J	https://www.youtube.com/watch?v=gALZRL4v2NM
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Y2 -- Find side lengths needed and calculate perimeter.		
Video	How to Find the Perimeter of a Rectangle Math with Mr. J	https://www.youtube.com/watch?v=teMDmMxy550
Video	How to Find Perimeter Math with Mr. J	https://www.youtube.com/watch?v=TSLc5p2VuYM
Worksheet (print)	Corbettmaths — Perimeter (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf
Y3 -- Compare perimeter and area of rectangles.		
Video	Same Area Different Perimeter Miss Hutchison	https://www.youtube.com/watch?v=bsl-r7dRSnc
Video	Finding the Area of a Rectangle Using the Formula Math with Mr. J	https://www.youtube.com/watch?v=1Ugk12y7TQM
Worksheet (print)	Corbettmaths — Perimeter (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf
Worksheet (print)	Corbettmaths — Area of a Rectangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf
Y4 -- Calculate elapsed time.		
Video	Elapsed Time — T-Chart Strategy Sally Osborne	https://www.youtube.com/watch?v=4Eucvb7qFpk
Video	Calculating Elapsed Time Using a Timeline EasyTeaching	https://www.youtube.com/watch?v=ML6r7BEZo7M
Worksheet (print)	Corbettmaths — Time Calculations (PDF)	https://corbettmaths.com/wp-content/uploads/2019/10/Time-Calculations-pdf.pdf

Type	Resource	Link
Y5 -- Read timetables and solve duration problems.		
Video	Timetables — Primary Corbettmaths	https://www.youtube.com/watch?v=40BJ0o0VD-0
Video	How to Read Timetables Maths Education EasyTeaching	https://www.youtube.com/watch?v=SZv_a0fjDfE
Worksheet (print)	Corbettmaths — Timetables (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Timetables-pdf.pdf
G1 -- Area and composite area.		
Video	Finding the Area of a Triangle — A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=-ZPfB1uDjEw
Video	How to Find the Area of a Composite Shape — L-Shaped Example Math with Mr. J	https://www.youtube.com/watch?v=LAXHgQwcrDc
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
G2 -- Distance, speed and time (rates).		
Video	Speed Distance Time Corbettmaths	https://www.youtube.com/watch?v=dHVK7leLGT8
Video	Speed Distance Time (Alternative Approach) Corbettmaths	https://www.youtube.com/watch?v=dOnpaBN5DAk
Worksheet (print)	Corbettmaths — Speed Distance Time (PDF)	https://corbettmaths.com/wp-content/uploads/2018/09/Speed-Distance-Time-pdf.pdf
G3 -- Interpret distance-time graphs.		
Video	Distance Time Graphs Corbettmaths	https://www.youtube.com/watch?v=oxOiuqUfWKA
Video	Speed Distance Time Corbettmaths	https://www.youtube.com/watch?v=dHVK7leLGT8
Worksheet (print)	Corbettmaths — Distance-Time Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/distance-time-graphs-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Finding the Area of a Triangle — A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=-ZPfB1uDjEw
Video	Speed Distance Time Corbettmaths	https://www.youtube.com/watch?v=dHVK7leLGT8
Video	Distance Time Graphs Corbettmaths	https://www.youtube.com/watch?v=oxOiuqUfWKA
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Worksheet (print)	Corbettmaths — Distance-Time Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/distance-time-graphs-pdf.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Metric Units (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf>
- ☐ Corbettmaths — Perimeter (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf>
- ☐ Corbettmaths Primary — Time (PDF) -- <https://corbettmathsprimary.com/wp-content/uploads/2018/07/time-pdf.pdf>
- ☐ Corbettmaths — Area of a Rectangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf>
- ☐ Corbettmaths — Time Calculations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2019/10/Time-Calculations-pdf.pdf>
- ☐ Corbettmaths — Timetables (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Timetables-pdf.pdf>
- ☐ Corbettmaths — Area of a Triangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf>
- ☐ Corbettmaths — Area of Compound Shapes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf>
- ☐ Corbettmaths — Speed Distance Time (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/09/Speed-Distance-Time-pdf.pdf>
- ☐ Corbettmaths — Distance-Time Graphs (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/distance-time-graphs-pdf.pdf>